

Column Name	Data Type	Description	Constraints
flight_fact_key	INT	Unique surrogate key for each flight fact record.	PK, NOT NULL
date_key	INT	Foreign key to the date dimension for the flight date.	FK dim_date.date_key, nullable
departure_airport_key	INT	Foreign key to the airport dimension that identifies the departure airport.	FK dim_airport.airport_key, nullable
arrival_airport_key	INT	Foreign key to the airport dimension that identifies the arrival airport.	FK dim_airport.airport_key, nullable
airline_key	INT	Foreign key to the airline dimension that identifies the operating airline.	FK dim_airline.airline_key, nullable
status_key	INT	Foreign key to the flight status dimension that identifies current/recorded flight status.	FK dim_flight_status.status_key, nullable
aircraft_key	INT	Foreign key to the aircraft dimension identifying the aircraft used for the flight.	FK dim_aircraft.aircraft_key, nullable
route_key	INT	Foreign key to the route dimension identifying the origin-destination route.	FK dim_route.route_key, nullable
source_file_name	VARCHAR	Name of the source file from which the flight record was ingested.	Nullable
flight_date	DATE	Calendar date associated with the flight record.	Nullable
flight_number	INT	Airline flight number used to identify the flight.	Nullable
flight_iata	VARCHAR	IATA-formatted flight code.	Nullable
flight_icao	VARCHAR	ICAO-formatted flight code.	Nullable
departure_terminal	INT	Departure terminal assigned to the flight.	Nullable
departure_gate	VARCHAR	Departure gate assigned to the flight.	Nullable
arrival_terminal	VARCHAR	Arrival terminal assigned to the flight.	Nullable
arrival_gate	VARCHAR	Arrival gate assigned to the flight.	Nullable
arrival_baggage	VARCHAR	Baggage claim area assigned for the arriving flight.	Nullable
departure_scheduled_ts	DATETIME	Scheduled departure timestamp.	Nullable, depends on source
departure_estimated_ts	DATETIME	Estimated departure timestamp after updates.	Nullable, depends on source
departure_actual_ts	DATETIME	Actual departure timestamp when available.	Nullable, depends on source
departure_estimated_runway_ts	DATETIME	Estimated runway departure timestamp.	Nullable, depends on source
departure_actual_runway_ts	DATETIME	Actual runway departure timestamp.	Nullable, depends on source
arrival_scheduled_ts	DATETIME	Scheduled arrival timestamp.	Nullable, depends on source
arrival_estimated_ts	DATETIME	Estimated arrival timestamp after updates.	Nullable, depends on source
arrival_actual_ts	DATETIME	Actual arrival timestamp when available.	Nullable, depends on source
arrival_estimated_runway_ts	DATETIME	Estimated runway arrival timestamp.	Nullable, depends on source
arrival_actual_runway_ts	DATETIME	Actual runway arrival timestamp.	Nullable, depends on source
departure_delay_minutes	INT	Number of minutes the flight departed later than scheduled.	Nullable unless required by source
arrival_delay_minutes	INT	Number of minutes the flight arrived later than scheduled.	Nullable unless required by source
is_codeshare	BOOLEAN / INT	Indicator showing whether the flight is operated as a codeshare.	Boolean
is_live_tracked	BOOLEAN / INT	Indicator showing whether live tracking data is available.	Boolean
is_cancelled	BOOLEAN / INT	Indicator showing whether the flight was cancelled.	Boolean
is_delayed	BOOLEAN / INT	Indicator showing whether the flight was delayed.	Boolean
is_landed	BOOLEAN / INT	Indicator showing whether the flight has landed.	Boolean
is_active	BOOLEAN / INT	Indicator showing whether the flight is currently active.	Boolean
is_scheduled	BOOLEAN / INT	Indicator showing whether the flight is scheduled.	Boolean
is_diverted	BOOLEAN / INT	Indicator showing whether the flight was diverted.	Boolean
flight_count	INT	Additive measure used to count one flight record in BI reports.	Nullable unless required by source
live_latitude	VARCHAR	Live aircraft latitude coordinate when tracking data is available.	Nullable unless required by source
live_longitude	VARCHAR	Live aircraft longitude coordinate when tracking data is available.	Nullable unless required by source

live_altitude	VARCHAR	Live aircraft altitude when tracking data is available.	Nullable unless required by source
live_direction	VARCHAR	Live aircraft direction or heading when tracking data is available.	Nullable unless required by source
live_speed_horizontal	VARCHAR	Live horizontal speed of the aircraft when tracking data is available.	Nullable unless required by source
live_speed_vertical	VARCHAR	Live vertical speed of the aircraft when tracking data is available.	Nullable unless required by source
live_is_ground	BOOLEAN / INT	Indicator showing whether the aircraft is on the ground in live tracking data.	Boolean
date_key	INT	Unique date key in YYYYMMDD format.	PK, NOT NULL
full_date	DATE	Full calendar date.	Nullable unless required by source
day_of_month	INT	Day number within the month.	Nullable unless required by source
day_of_week	INT	Numeric day of week.	Nullable unless required by source
day_name	VARCHAR	Name of the weekday.	Nullable unless required by source
week_of_year	INT	Week number within the year.	Nullable unless required by source
month_number	INT	Month number.	Nullable unless required by source
month_name	VARCHAR	Month name.	Nullable unless required by source
quarter_number	INT	Calendar quarter number.	Nullable unless required by source
year_number	INT	Calendar year.	Nullable unless required by source
is_weekend	BOOLEAN / INT	Indicator showing whether the date falls on a weekend.	Boolean
airport_key	INT	Unique surrogate key for each airport dimension record.	PK, NOT NULL
airport_id	INT	Source system identifier for the airport.	Nullable unless required by source
iata_code	VARCHAR	IATA code used to identify the airline or airport.	Code value that is nullable
icao_code	VARCHAR	ICAO code used to identify the airline or airport.	Code value that is nullable
airport_name	VARCHAR	Official or common name of the airport.	Nullable unless required by source
city_iata_code	VARCHAR	IATA city code associated with the airport.	Code value that is nullable
city_name	VARCHAR	City served by the airport.	Nullable unless required by source
country_iso2	VARCHAR	Two-letter ISO country code.	Nullable unless required by source
country_name	VARCHAR	Country name associated with the record.	Nullable unless required by source
continent	VARCHAR	Continent where the airport is located.	Nullable unless required by source
timezone	VARCHAR	Timezone of the airport.	Nullable unless required by source
gmt	VARCHAR	GMT offset for the airport timezone.	Nullable unless required by source
latitude	DECIMAL	Latitude coordinate of the airport.	Nullable unless required by source
longitude	DECIMAL	Longitude coordinate of the airport.	Nullable unless required by source
geoname_id	INT	GeoNames reference identifier for the airport location.	Nullable unless required by source
phone_number	VARCHAR	Airport contact phone number when provided.	Nullable unless required by source
airline_key	INT	Unique surrogate key for each airline dimension record.	PK, NOT NULL, UNIQUE
airline_id	INT	Source system identifier for the airline.	Nullable unless required by source/schema
iata_code	VARCHAR	IATA code used to identify the airline or airport.	Code value that is nullable
icao_code	VARCHAR	ICAO code used to identify the airline or airport.	Code value that is nullable
airline_name	VARCHAR	Official or common name of the airline.	Nullable unless required by source
callsign	VARCHAR	Radio callsign used by the airline.	Nullable unless required by source
hub_code	VARCHAR	Primary hub airport code for the airline.	Code that is nullable if the source is missing
country_iso2	VARCHAR	Two-letter ISO country code.	Nullable unless required by source
country_name	VARCHAR	Country name associated with the record.	Nullable unless required by source
date_founded	DATE	Date or year the airline was founded, when available.	Nullable unless required by source
fleet_size	INT	Number of aircraft in the airline fleet.	Nullable unless required by source
fleet_average_age	DECIMAL	Average age of the airline fleet.	Nullable unless required by source
iata_prefix_accounting	INT	IATA accounting prefix assigned to the airline.	Nullable unless required by source
airline_status	VARCHAR	Operational status of the airline.	Nullable unless required by source
airline_type	VARCHAR	Type/category of airline service.	Nullable unless required by source

aircraft_key	INT	Unique surrogate key for each aircraft dimension record.	PK, NOT NULL, UNIQUE
iata_type_code	VARCHAR	IATA aircraft type code.	Code that is nullable if the source is missing
aircraft_registration	VARCHAR	Aircraft registration/tail number.	Nullable unless its required by the source
aircraft_type_ref_id	INT	Reference identifier for the aircraft type from the source system.	Nullable unless its required by the source
plane_type_id	INT	Source identifier for the plane type.	Nullable unless its required by the source
aircraft_name	VARCHAR	Name or label of the aircraft when provided by the source.	Nullable unless its required by the source
iata_code_long	VARCHAR	Long IATA aircraft code.	Nullable unless its required by the source
iata_code_short	VARCHAR	Short IATA aircraft code.	Nullable unless its required by the source
model_code	VARCHAR	Aircraft model code.	Code that is nullable if the source is missing
model_name	VARCHAR	Aircraft model name.	Nullable unless required by source
plane_series	VARCHAR	Aircraft series/family classification.	Nullable unless required by source
plane_class	INT	Aircraft class/category.	Nullable unless required by source
production_line	VARCHAR	Aircraft production line or manufacturer family.	Nullable unless required by source
engines_count	INT	Number of engines on the aircraft.	Nullable unless required by source
engines_type	VARCHAR	Type of engines used by the aircraft.	Nullable unless required by source
plane_status	VARCHAR	Status of the aircraft record, such as active or historical.	Nullable unless required by source
route_key	INT	Unique surrogate key for each route dimension record.	PK, NOT NULL,
origin_iata_code	VARCHAR	IATA code for the origin airport.	value that is nullable
origin_icao_code	VARCHAR	ICAO code for the origin airport.	value that is nullable
destination_iata_code	VARCHAR	IATA code for the destination airport.	value that is nullable
destination_icao_code	VARCHAR	ICAO code for the destination airport.	value that is nullable
route_name	VARCHAR	Readable route name combining origin and destination airport codes.	Nullable unless required by source
is_domestic	BOOLEAN / INT	Indicator showing whether the route is within one country.	Boolean
is_international	BOOLEAN / INT	Indicator showing whether the route crosses countries.	Boolean
origin_country_name	VARCHAR	Country name of the origin airport.	Nullable unless required by source
destination_country_name	VARCHAR	Country name of the destination airport.	Nullable unless required by source
origin_timezone	VARCHAR	Timezone of the origin airport.	Nullable unless required by source
destination_timezone	VARCHAR	Timezone of the destination airport.	Nullable unless required by source
status_key	INT	Unique surrogate key for each flight status record.	PK and NOT NULL
flight_status	VARCHAR	Descriptive flight status value from the source system.	Nullable unless required by source
is_cancelled_flag	BOOLEAN / INT	Flag identifying cancelled status records.	Boolean
is_delayed_flag	BOOLEAN / INT	Flag identifying delayed status records.	Boolean
is_landed_flag	BOOLEAN / INT	Flag identifying landed status records.	Boolean
is_active_flag	BOOLEAN / INT	Flag identifying active status records.	Boolean
is_scheduled_flag	BOOLEAN / INT	Flag identifying scheduled status records.	Boolean
is_diverted_flag	BOOLEAN / INT	Flag identifying diverted status records.	Boolean